

VLAN Hopping Attacks

CompTIA PenTest+ (PT0-003)

Domain: 4.0: Attacks and Exploits

Objective: 4.2: Given a scenario, perform network attacks using the appropriate tools.

Video Overview

Welcome to this video on VLAN Hopping Attacks! As you prepare for your PenTest+ certification, understanding these advanced attack techniques is essential for developing a comprehensive cybersecurity skillset. We will explore how attackers can bypass network segmentation to gain unauthorized access to sensitive Virtual Local Area Networks (VLANs), a concept important in real-world network security. This knowledge is directly applicable to a career in penetration testing or network defense, enabling you to identify vulnerabilities and implement effective mitigation strategies in diverse professional environments.

Key Topics Covered

- **VLAN Hopping:** Techniques allowing an attacker to bypass network segmentation to gain access to unauthorized VLANs.
- **Network Segmentation:** A defense strategy that isolates different parts of a network to limit attacker access.
- **Switch Spoofing:** An attack method where the attacker's machine pretends to be a legitimate network switch to exploit trunking mode defaults.
- **Dynamic Trunking Protocol (DTP):** Packets used in switch spoofing to tell a legitimate switch that the attacker's machine is a trunk port.
- **Double Tagging:** A technique that encapsulates packets with two VLAN tags (inner for desired VLAN, outer for native VLAN) to bypass segmentation.
- **Scapy:** A tool effective for configuring and sending custom packets for switch spoofing and double tagging attacks.
- **Wireshark:** A packet analyzer used to observe network traffic and confirm the success of VLAN hopping attacks.
- **Mitigation Strategies:** Understanding how newer equipment and proper configuration best practices can reduce the risk of VLAN hopping attacks.

Student Notes Section